Stable Matching and School Choice

Students rank schools; schools rank students by priority. Under immediate acceptance, a student may benefit from ranking a safer school first. Deferred acceptance changes when an assignment becomes final.

Definition 1. A *matching market* consists of a set A of n students and a set B of n schools, where each student a has a strict, complete preference order $\succ_a$ over B, and each school b has a strict, complete priority order $\succ_b$ over A. A *matching* M is a bijection between A and B; we write $M(a)$ for a's partner and $M(b)$ for b's partner.

Definition 2. A pair (a, b) with $M(a) \neq b$ is a *blocking pair* for the matching M if

$$b \succ_a M(a) \quad \text{and} \quad a \succ_b M(b),$$

that is, if a prefers b to its assigned partner and b prefers a to their assigned partner. A matching with no blocking pair is *stable*.

What does a blocking pair allow its two members to do?

Both strictly prefer to match with each other rather than keep their assigned partners.

Deferred Acceptance

Other uses of DA. In the NRMP residency match, applicants propose to programs in rank order and programs hold tentative matches. New York City's 2003–04 high school redesign also used a version of DA with students proposing.[1]

We start with students as proposers and schools as receivers. In the one-seat model we can also reverse the roles, with schools proposing in priority order and students holding their favorite offers. A proposal here is a step in the algorithm. The district sends the final offers.

Build the rule

Why hold the application tentatively? A higher-priority student may apply later. Holding a keeps a seat available for them if no higher-priority student applies.
How should a school handle a new application? Keep the student with higher priority. The student turned down applies to their next school.
When can we finalize? When no unheld student can make another proposal. We will show that in this model every student is then held.

[1] NRMP, *How It Works*, `nrmp.org`. Abdulkadiroglu, Pathak, and Roth, NBER Working Paper 14864.

Deferred Acceptance (DA), student-proposing

a. While some student a is unheld and has not yet proposed to every school:

(i) a proposes to the most-preferred school b on a's list among those not yet proposed to.

(ii) If b is holding nobody, then b *tentatively holds* a.

(iii) Otherwise b holds some a'; they keep whichever of a, a' they prefer and reject the other, who becomes unheld.

b. Output the matching M in which each school is assigned the student it holds.

Theorem 1. *Deferred acceptance terminates after at most n^2 proposals, and the matching it outputs is a stable perfect matching. In particular, a stable matching always exists.*

Proof. Each student moves down their list without repeating a proposal, so the algorithm terminates after at most n^2 proposals. Once a school holds a student, it stays occupied and only switches to students with higher priority. If an unheld student had exhausted their list, all n schools would hold distinct students, including that student. Thus the algorithm cannot stop with an unheld student, and the output is perfect.

Consider a pair (a, b) outside the output. If a proposed to b, then b rejected a immediately or later in favor of a student with higher priority. Because schools only trade up, $M(b) \succ_b a$. If a never proposed to b, then a stopped at a school higher on their list, so $M(a) \succ_a b$. Neither case permits a blocking pair. □

Sanity check: how many proposals are made if every student has a different first choice?

Exactly n. Every proposal is held and nobody is rejected.

Exercise. Run deferred acceptance with students proposing and then with schools proposing. Compare the partners.

$$a_1 : b_1 \succ b_2 \qquad b_1 : a_2 \succ a_1$$
$$a_2 : b_2 \succ b_1 \qquad b_2 : a_1 \succ a_2$$

The two outputs.

$$M_A = \{(a_1, b_1), (a_2, b_2)\}, \qquad M_B = \{(a_1, b_2), (a_2, b_1)\}.$$

In each run, everyone on the proposing side receives their first choice. Both outputs are stable: in M_A no a wants to deviate; in M_B no b wants to deviate.

Student Optimality

Definition 3. School b is *achievable* for student a if there is some stable matching M with $M(a) = b$.

Theorem 2 (Gale–Shapley '62)**.** *Student-proposing deferred acceptance matches every student to their most-preferred achievable school. In particular, all students simultaneously get the best partner they could get in any stable matching, so there is a single* student-optimal *stable matching.*

Claim 1. Over the course of the algorithm, no student is ever rejected by an achievable school.

Why does the claim imply student optimality?

Fix a and let $b^* = M(a)$ be their DA school. Students walk down their lists in order, so every school a prefers to b^* rejected a, and by the claim each of those is unachievable for a. And b^* is achievable, since Theorem 1 says DA's output is stable. So b^* is the top achievable school on a's list.

Proof of Claim 1. **Setup.** Suppose not, and take the very first rejection in the run in which some a is rejected by a school b achievable for a. By (iii), b rejected a in favor of some a' with $a' \succ_b a$. Fix a stable M with $M(a) = b$ and let $b' = M(a')$; note $b' \neq b$, since b is matched to a in M.

What a' has been doing. At that moment a' has proposed to b, so they have already been rejected by every school they prefers to b. All of those rejections happened *before* the first rejection by an achievable school, so all of those schools are unachievable for a'. But b' is achievable for a'—witness M. So b' isn't one of them, and since $b' \neq b$ we get $b \succ_{a'} b'$.

The contradiction. So a' prefers b to $M(a') = b'$, and b prefers a' to $M(b) = a$. That is precisely tthey statement that (a', b) blocks M. □

Exercise. Where does the proof use that this is the first rejection by an achievable school?

Every school that previously rejected a' is unachievable for a'. Since $M(a')$ is achievable, a' cannot prefer $M(a')$ to b.

School Choice

Students report preference lists. School priorities are fixed strict orders. A student–school pair blocks an assignment if the student prefers that school and it has a vacancy or an assigned student with lower priority. Schools may have several seats. With fixed strict priorities, a school tentatively holds its highest-priority applicants up to capacity and rejects the rest.

Theorem 3. *Student-proposing deferred acceptance is strategyproof for students: reporting a true preference list is a dominant strategy.*

Student optimality alone does not prove strategyproofness: a matching stable under a misreport need not be stable under the true preferences.

In the *Boston mechanism*, acceptances are final. In round k, each unassigned student applies to the school ranked kth; each school fills its remaining seats in priority order.
Immediate acceptance in practice. Boston used this rule before its 2005 decision to switch to DA. Chicago also used a version for selective high school admissions before its 2009 reform.[2]

Exercise. Each school has one seat. Run immediate acceptance on these preferences. Can one student obtain a better school by changing their report?

Student	1st	2nd	3rd	Priority at	1st	2nd	3rd
s_1	Y	X	Z	X	s_1	s_3	s_2
s_2	X	Y	Z	Y	s_2	s_1	s_3
s_3	X	Y	Z	Z	s_3	s_2	s_1

Everyone reports truthfully. Round 1: Y sees only s_1 and admits them; X sees s_2, s_3 and admits s_3, since $s_3 \succ_X s_2$. Round 2: s_2 applies to their second choice Y—full, permanently. Round 3: s_2 takes Z.

$$s_1 \mapsto Y, \qquad s_2 \mapsto Z, \qquad s_3 \mapsto X.$$

So s_2, reporting honestly, lands their last choice.

s_2 **misreports** $Y \succ X \succ Z$**.** Round 1: Y now sees s_1 and s_2 and admits s_2, since $s_2 \succ_Y s_1$; X sees only s_3 and admits them. Round 2: $s_1 \to X$, full. Round 3: s_1 takes Z.

$$s_1 \mapsto Z, \qquad s_2 \mapsto Y, \qquad s_3 \mapsto X.$$

Student s_2 has gone from Z to Y by lying, so truth-telling is not a dominant strategy.
Is the truthful Boston outcome stable? What does student-proposing DA return?

The pair (s_2, Y) blocks: s_2 prefers Y to Z, and Y gives s_2 priority over s_1. DA returns $s_1 \mapsto X$, $s_2 \mapsto Y$, $s_3 \mapsto Z$.

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References

Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory.* Cambridge University Press, 2016.

[2] Abdulkadiroglu et al., NBER Working Paper 11965. Pathak and Sönmez, NBER Working Paper 16783.